Appendix A

Datasets and APIs for Financial AI

A.1 Introduction

Every chapter in this book is built around a worked numerical example, and several of the later chapters, most notably Chapter 18's UCI credit-card default case study, are built around a real, publicly available dataset rather than a fictional one. This appendix collects, in one place, the data sources a reader is most likely to reach for when reproducing this book's examples with real data or extending them into an independent project: what each source covers, how it is typically licensed or priced, and the specific API or library used to pull it into Python.

This appendix is deliberately organized around *access mechanism* as much as content, since the practical difference between a dataset a reader can query in five minutes with a free API key and one that requires an institutional subscription is often the difference between a project that gets started and one that does not. Where this book's own companion notebooks already reach outside the book's fictional running examples for real data, chiefly Chapter 5's CBOE Volatility Index (VIX) history (Section A.2, downloaded live via the `yfinance` library) and Chapter 18's UCI "Default of Credit Card Clients" dataset (Section A.6), this appendix cross-references the relevant chapter directly.

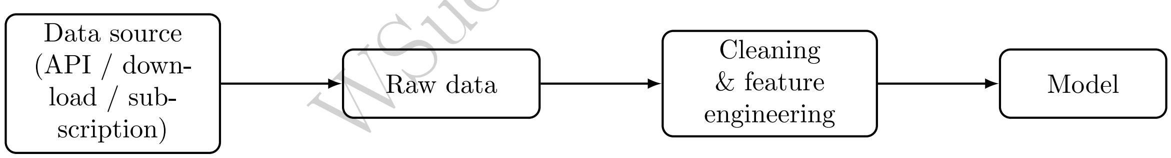

Figure A.1: Where the datasets and APIs in this appendix sit in a typical workflow: they supply the raw data, but the cleaning and feature-engineering steps introduced in Chapter 2's data-cleaning section and used throughout this book still have to be applied before the data reaches a model.

A.2 Market Price and Trading Data

Table A.1 summarizes the most commonly used sources for equity, index, foreign exchange, and derivatives price data. Yahoo Finance, accessed through the `yfinance` Python library, is the source this book itself uses: Chapter 5's implied-volatility section downloads the actual daily history of the VIX this way, with no API key required. It is an unofficial wrapper around Yahoo Finance's public website rather than a documented, versioned API, which is precisely why it is free; the tradeoff, discussed further in Section A.13, is that it can change behavior or break without notice whenever Yahoo changes its own site.

Where CRSP and Compustat are US-centric, Refinitiv Datastream (a product of the London Stock Exchange Group since its 2021 acquisition of Refinitiv) is the closest institutional equivalent for international and cross-country work: decades of equity, index, bond, and FX price history across essentially every major market, not just the US, together with the Worldscope database of global company fundamentals bundled into the same platform. Like WRDS, it is licensed rather than freely available, typically through a university or employer subscription, and is accessed either through its legacy Excel add-in or through the Datastream Web Service (DSWS), a REST API with an official `DatastreamPy` Python client.

Table A.1: Market price and trading data sources.

Source	Access	API / library	Notes
Yahoo Finance	Free	`yfinance` (PyPI)	Equities, ETFs, indices, FX, crypto, options chains; used in Chapters 2 and 5 of this book; unofficial, can break without notice
Alpha Vantage	Free tier + paid	REST API, key required (`alphavantage.co`)	Equities, FX, crypto, and technical indicators; free tier is rate-limited to a handful of calls per minute
Nasdaq Data Link (formerly Quandl)	Free + paid tiers	REST API (`nasdaqdatalink.com`)	Broad marketplace of market, commodity, and economic datasets from many contributors
Polygon.io	Free limited tier + paid	REST and WebSocket API	Real-time and historical US equities, options, FX, and crypto, including tick-level data
Tiingo	Free tier + paid	REST and WebSocket API	End-of-day prices, fundamentals, and news
CBOE DataShop	Paid	Web portal, bulk download	Official source for VIX methodology data and listed options data
WRDS: CRSP, TAQ	Institutional subscription (typically via a university)	Web query tool, plus SAS/Python/R client	CRSP is the academic gold standard for US equity returns; TAQ provides tick-by-tick trade and quote data used in Chapter 12's microstructure examples
Refinitiv Datastream (LSEG)	Institutional subscription (typically via a university or employer)	Excel add-in, or the Datastream Web Service (DSWS) REST API via `DatastreamPy`	The standard source for international/cross-country equity, index, bond, and FX history, plus Worldscope global fundamentals in the same platform; fills the gap CRSP and Compustat leave outside the US

A.3 Options and Derivatives Data

Options data deserves its own treatment because it is structurally different from equity price data: a single underlying can have hundreds of simultaneously listed contracts across many strikes and expirations, and the quantity most useful for the work in Chapter 5 (pricing, the Greeks, implied volatility) and Chapter 8 (delta-normal and delta-gamma VaR for an options position) is not the raw quoted price but the implied volatility and Greeks computed from it, which not every source provides pre-computed. OptionMetrics, accessed through WRDS as the IvyDB US database, is the main dataset academic options research is built on, occupying essentially the same role for options that CRSP occupies for equities (Section A.2): daily, exchange-sourced, precomputed implied volatilities and Greeks for every listed US equity and index option going back to 1996, standardized enough that a result computed on it is directly comparable across studies. It is not free, licensed only through an institutional WRDS subscription, which is precisely why Table A.2 also lists several lighter-weight alternatives, from a free but shallow current-chain snapshot to paid practitioner-oriented vendors, for a reader without WRDS access.

A.4 Fundamental and Financial Statement Data

The U.S. Securities and Exchange Commission's EDGAR system is the single most important free source of fundamental data for U.S. public companies, and it is directly relevant to Chapter 4's treatment of the income statement, balance sheet, and cash flow statement. Beyond the human-readable filings, the SEC exposes two machine-readable APIs at `data.sec.gov`: a full-text search API for locating specific language across filings (directly usable for Chapter 14's 10-K risk-factor classification example), and an XBRL "company facts" API that returns every structured financial figure, revenue, COGS, CFO, and so on, that a company has ever reported, tagged and time-stamped, for a given ticker or CIK (Central Index Key). Because these figures come directly from a company's own regulatory filings, they require no scraping or vendor licensing, only a descriptive `User-Agent` header identifying the requester, which the SEC requires of all automated EDGAR access.

A.5 Macroeconomic and Interest Rate Data

FRED, the Federal Reserve Bank of St. Louis's Federal Reserve Economic Data service, is the standard free source for the interest-rate and macroeconomic series this book uses conceptually in Chapter 5's term-structure discussion and Chapter 8's stress-testing scenarios. A free API key registered at `fred.stlouisfed.org` grants access to its REST API, and the community-maintained `fredapi` Python package wraps it in a single function call that returns a time-indexed `pandas` series, matching the data structures this book's notebooks already use throughout.

A.6 Credit, Lending, and Consumer Finance Data

This is the category with the deepest overlap with this book's own worked examples. The UCI "Default of Credit Card Clients" dataset is already Chapter 18's own running case study, downloaded programmatically by that chapter's notebook rather than committed to this repository. The other datasets in Table A.5 are natural extensions for the capstone project ideas Chapter 18 closes with.

A.7 Fraud, AML, and Regulatory Compliance Data

Genuinely labeled fraud data is scarce, since financial institutions rarely release real transaction-level fraud labels, which is exactly why the two datasets in Table A.6 recur so often in both industry benchmarks and this book's own Chapter 13 and Chapter 18 capstone list.

A.8 Text, News, and Sentiment Data

Chapter 14's treatment of sentiment scoring rests on the Loughran-McDonald Master Dictionary, a word list purpose-built for financial text (ordinary sentiment dictionaries misclassify routine financial vocabulary, such as "liability" or "tax," as negative), freely downloadable from its maintainer's University of Notre Dame page. Table A.7 lists this and the other text sources most relevant to that chapter.

Table A.2: Options and derivatives data sources.

Source	Access	API / library	Notes
Yahoo Finance	Free	`yfinance`'s `Ticker.option_chain()`	Current listed option chains (bid, ask, volume, open interest) for a given expiration; no deep history, and implied volatility is Yahoo's own estimate rather than a modeled surface
CBOE DataShop	Paid	Web portal, bulk download	Official historical end-of-day and intraday options quotes directly from the exchange that lists most US index and equity options
Polygon.io Options API	Free limited tier + paid	REST and WebSocket API	Historical and real-time options chains with server-computed Greeks and implied volatility
WRDS: OptionMetrics (IvyDB US)	Institutional subscription	Web query tool, plus SAS/Python/R client	The academic standard for US options research: daily implied volatility surfaces and Greeks, precomputed, going back to 1996; directly usable for Chapters 5 and 8
ORATS	Paid	REST API	Historical implied volatility surfaces, Greeks, and options analytics aimed at practitioners rather than academics
Deribit	Free, public endpoints	REST and WebSocket API, no key required for public data	The largest crypto options venue; full order book and historical settlement data for BTC and ETH options, useful for practicing the same Black-Scholes-Merton and Greeks machinery from Chapter 5 on an asset class with far higher realized volatility
CME DataMine	Paid	Bulk download / API	Historical futures and futures-options data (rates, commodities, equity index futures) directly from the exchange

Table A.3: Fundamental and financial statement data sources.

Source	Access	API / library	Notes
SEC EDGAR	Free	REST API (`data.sec.gov`); full-text search and XBRL company-facts endpoints	Every US public company's own filings; ties directly to Chapters 4 and 14
Financial Modeling Prep	Free tier + paid	REST API	Pre-parsed income statements, balance sheets, cash flow statements, and ratios
WRDS: Compustat	Institutional subscription	Web query tool, plus SAS/Python/R client	Standardized, restated fundamentals used throughout academic finance research
Macrotrends, StockAnalysis.com	Free (web only)	No official API	Convenient for spot-checking a single company's numbers by hand; not suitable for programmatic bulk use

Table A.4: Macroeconomic and interest rate data sources.

Source	Access	API / library	Notes
FRED	Free, key required	REST API (`fred.stlouisfed.org`); `fredapi` package	Treasury yields, GDP, inflation, unemployment; directly usable for Chapters 5 and 8
World Bank Open Data	Free	REST API (`worldbank.org`)	Global macro and development indicators across countries
IMF Data	Free	REST API	Global macro and financial-stability indicators
U.S. Bureau of Labor Statistics	Free, key optional	REST API (`bls.gov`)	US employment and price-inflation series

Table A.5: Credit, lending, and consumer finance data sources.

Source	Access	API / library	Notes
UCI Default of Credit Card Clients	Free	Direct download via `urllib`	Chapter 18's own dataset; Taiwanese credit-card default data
UCI German Credit Data	Free	Direct download	Small, classic credit-scoring dataset, conceptually close to Chapter 6 and 9's examples
Lending Club loan data	Free (via Kaggle mirror)	Kaggle API	Loan-level peer-to-peer lending data; a Chapter 18 capstone idea
HMDA (Home Mortgage Disclosure Act) data	Free	REST API via the Consumer Financial Protection Bureau's data browser (`consumerfinance.gov`)	Mortgage application-level data widely used for fair-lending analysis; directly usable for Chapter 16's fairness audits

Table A.6: Fraud, AML, and regulatory compliance data sources.

Source	Access	API / library	Notes
IEEE-CIS Fraud Detection (Kaggle)	Free (Kaggle competition dataset)	Kaggle API	Real, anonymized e-commerce transaction fraud data; a Chapter 13/17 capstone idea
PaySim	Free	Kaggle API	Synthetic mobile-money transaction data modeled on a real provider's logs; well suited to practicing the structuring and layering detection introduced in Chapter 13
OFAC Sanctions Lists	Free	Bulk data files and a REST search API via the US Treasury	Sanctioned-party screening lists used in AML compliance workflows

Table A.7: Text, news, and sentiment data sources.

Source	Access	API / library	Notes
SEC EDGAR full-text search	Free	REST API (`data.sec.gov`)	Search across filing text; directly usable for Chapter 14's 10-K risk-factor classification example
Loughran-McDonald Master Dictionary	Free	Direct CSV/text download	The sentiment word list underlying Chapter 14's scoring examples
Financial PhraseBank	Free	Hugging Face `datasets` library, or direct download	Sentence-level, analyst-labeled financial news sentiment
GDELT Project	Free	REST API and BigQuery access	Global news event and tone database, updated continuously
Reddit API	Free, registered app required, rate-limited	`praw` (Python Reddit API Wrapper)	Social sentiment data, e.g., retail-investor forums
X (formerly Twitter) API	Paid tiers	REST and streaming API	Social sentiment; access has moved to paid tiers since 2023, unlike the largely free access assumed by older tutorials

A.9 Alternative Data

Chapter 15's case vignette on satellite-derived retail and commodity signals (RS Metrics, UBS, and Orbital Insight's oil-storage monitoring) is representative of this entire category: the underlying vendor data is almost always enterprise-priced and licensed rather than available through a simple API key, since it is expensive to produce and its main customers are institutional investors, not students. Table A.8 separates the handful of alternative-data sources that are genuinely free or low-cost from the enterprise vendors that are not, so a reader does not spend time chasing a free tier that does not exist.

Table A.8: Alternative data sources.

Source	Access	API / library	Notes
Google Trends	Free, unofficial	`pytrends` (unofficial wrapper)	Search-interest series, often used as an attention or sentiment proxy
Orbital Insight, RS Metrics	Enterprise, paid	Vendor portal/API	Satellite-derived alternative data referenced in Chapter 15's case vignette; not accessible without a commercial contract
Similarweb, Sensor Tower	Paid	Vendor API	Web-traffic and app-usage panel data
MSCI ESG, Sustainalytics, Refinitiv ESG	Paid	Vendor API	ESG scores and underlying metrics

A.10 High-Frequency and Market Microstructure Data

Reconstructing an actual limit order book, the exercise Chapter 3 introduces by hand and Chapters 11 and 12 build trading strategies on top of, requires message-level data that most retail-facing APIs simply do not provide. LOBSTER, an academic data service built on top of NASDAQ's own historical ITCH data, is the most accessible source of genuine, reconstructed limit-order-book data: it offers a free academic tier (typically a handful of sample days for a limited set of tickers) alongside paid access to its full history.

A.11 Factor, Index, and Portfolio Data

No source is more central to Chapter 10's factor-model examples than the Ken French Data Library, maintained by Dartmouth's Tuck School of Business, which has published the Fama-French factor returns (market, size, value, and their later extensions) and a large set of industry-sorted portfolios as free, direct-download CSV and ZIP files for over two decades. It has no formal REST API, but its file layout and URLs have been stable for so long that the `pandas-datareader` package includes a dedicated reader for it.

A.12 Putting It Together: A Chapter-by-Chapter Dataset Map

Table A.11 condenses the sections above into a single reference: for each chapter of this book, which of the sources above is the most natural place to look for real data to extend that chapter's fictional examples.

Table A.9: High-frequency and market microstructure data sources.

Source	Access	API / library	Notes
LOBSTER	Free academic tier + paid	Web portal, bulk download (`lobsterdata.com`)	Reconstructed limit order book data for NASDAQ-listed stocks; directly usable for Chapters 3, 11, and 12
Databento	Pay-as-you-go	REST API, historical and live feeds	Modern, lower-cost alternative to legacy tick-data vendors for US equities, futures, and options
WRDS: NYSE TAQ	Institutional subscription	Web query tool, plus SAS/Python/R client	Tick-by-tick trade and quote data; the traditional academic source for Chapter 12-style microstructure research

Table A.10: Factor, index, and portfolio data sources.

Source	Access	API / library	Notes
Ken French Data Library	Free	Direct CSV/ZIP download; also wrapped by `pandas-datareader`	Fama-French factor returns and industry portfolios; directly usable for Chapter 10
iShares, SPDR ETF holdings	Free	Daily CSV download from the issuer's own website	Actual ETF basket composition; usable for Chapter 12's ETF-basket arbitrage example
S&P Dow Jones Indices	Free (delayed) + paid (full)	Vendor portal	Index membership, methodology documents, and historical levels

Chapter 18's own closing list of capstone project ideas already names several of these sources directly; this table extends that same mapping to every other chapter.

A.13 Practical Challenges in Sourcing Financial Data

Pulling real data into a project is rarely as simple as calling an API and moving on; several recurring problems are worth flagging explicitly.

Point-in-time and look-ahead bias. Most free APIs return a security's or company's data as it is known *today*, not as it was actually known on the date being studied. Fundamentals get restated after the fact, index membership changes, and a company's own name or ticker can change; naively joining "current" fundamentals to historical prices silently leaks future information into a backtest, exactly the look-ahead-bias problem Chapter 15 raises directly.

Survivorship bias. A free API that only lists currently-traded tickers implicitly excludes every company that has since been delisted, gone bankrupt, or been acquired, which biases any historical universe built from it toward survivors. CRSP's paid, point-in-time data is the academic standard specifically because it avoids this.

Licensing and redistribution restrictions. Most vendor terms of service prohibit redistributing raw data even for non-commercial or educational use, which is exactly why this book's own real-dataset chapters (Chapter 18, and this appendix's own Chapter 5 VIX example) download data programmatically at run time rather than committing it to the repository. The same discipline is worth following in any derivative project.

API and vendor instability. Unofficial APIs such as `yfinance` work by interacting with a public website rather than a documented, contractually stable interface, so they can change behavior or stop working without notice whenever the underlying site changes. Even official free tiers are periodically discontinued or replaced outright, as has happened repeatedly across the market-data-vendor landscape. Every specific vendor and endpoint named in this appendix should therefore be treated as illustrative rather than permanent: verify current availability and terms directly with the provider before building anything on top of one.

Rate limits and cost tiers. Free tiers are typically rate-limited (a fixed number of calls per minute or per day), which matters when a project needs to pull data for hundreds of tickers rather than one; budgeting for a paid tier, or batching and caching requests locally, is often unavoidable at scale.

A.14 Further Reading

- U.S. Securities and Exchange Commission, *EDGAR Application Programming Interfaces*, `sec.gov`.
- Federal Reserve Bank of St. Louis, *FRED API Documentation*, `fred.stlouisfed.org`.
- Kenneth R. French, *Data Library*, Tuck School of Business, Dartmouth College, `mba.tuck.dartmouth.edu`.
- LOBSTER, *Limit Order Book Data Documentation*, `lobsterdata.com`.
- Kaggle, *Datasets and Public API Documentation*, `kaggle.com`.
- UCI Machine Learning Repository, `archive.ics.uci.edu`.
- Loughran, T. and McDonald, B., "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks," *Journal of Finance*, 2011 (the paper underlying the Loughran-McDonald Master Dictionary).

Table A.11: A chapter-by-chapter map from this book's topics to the datasets and APIs most relevant to them.

Chapter	Most relevant data sources
1. Introduction	(conceptual; no dataset needed)
2. Python for Finance	Yahoo Finance / `yfinance`, for real practice alongside the chapter's synthetic AssetA/AssetB series
3. Financial Markets and Institutions	LOBSTER (limit order book reconstruction), Yahoo Finance
4. Financial Statements	SEC EDGAR (filings and XBRL company facts)
5. Valuation and Asset Pricing	FRED (yield curve), Yahoo Finance / `yfinance` (already used for the VIX in Section A.2), OptionMetrics or Deribit for real options chains (Section A.3)
6. Machine Learning Fundamentals	UCI German Credit Data, or any dataset above used generically for classification practice
7. Time Series Analysis and Forecasting	FRED (macro series), Yahoo Finance (price series)
8. Market Risk Management	Yahoo Finance and FRED, for a real-portfolio VaR backtest; OptionMetrics or Polygon.io for a real delta-normal/delta-gamma VaR options example (Section A.3)
9. Credit Risk Modeling	UCI Default of Credit Card Clients, Lending Club loan data, HMDA data
10. Portfolio Theory and Asset Allocation	Ken French Data Library, Yahoo Finance
11. Algorithmic Trading and Market Microstructure	LOBSTER, Yahoo Finance intraday data (where available)
12. High-Frequency Trading	LOBSTER, Databento, WRDS NYSE TAQ
13. Fraud Detection, Compliance, and Financial Crime	IEEE-CIS Fraud Detection, PaySim, OFAC sanctions lists
14. Natural Language Processing in Finance	SEC EDGAR full-text search, Loughran-McDonald dictionary, Financial PhraseBank, GDELT, Reddit API
15. Alternative Data and Deep Learning in Finance	Google Trends, satellite/alt-data vendors (enterprise), Similarweb
16. Explainability, Fairness, and Regulation	HMDA data, UCI Default of Credit Card Clients (Chapter 16's own baseline model)
17. Reinforcement Learning in Finance	Simulated market environments by design; LOBSTER or Yahoo Finance intraday data for a real extension of the chapter's execution and market-making agents
18. Case Studies and Capstone Projects	The full list above; see Chapter 18's own closing section for specific project pairings